

[illegible]

ESP32-WROVER-IE MODULE

ESPRESSIF

U8

Pinout details:

- Power:**
 - Pin 1: +3.3V
 - Pin 2: GND
 - Pin 3: +3.3V
 - Pin 4: GND
 - Pin 5: +3.3V
 - Pin 6: GND
 - Pin 7: +3.3V
 - Pin 8: GND
 - Pin 9: +3.3V
 - Pin 10: GND
 - Pin 11: +3.3V
 - Pin 12: GND
 - Pin 13: +3.3V
 - Pin 14: GND
 - Pin 15: +3.3V
 - Pin 16: GND
 - Pin 17: +3.3V
 - Pin 18: GND
 - Pin 19: +3.3V
 - Pin 20: GND
 - Pin 21: +3.3V
 - Pin 22: GND
 - Pin 23: +3.3V
 - Pin 24: GND
 - Pin 25: +3.3V
 - Pin 26: GND
 - Pin 27: +3.3V
 - Pin 28: GND
 - Pin 29: +3.3V
 - Pin 30: GND
 - Pin 31: +3.3V
 - Pin 32: GND
 - Pin 33: +3.3V
 - Pin 34: GND
 - Pin 35: +3.3V
 - Pin 36: GND
 - Pin 37: +3.3V
 - Pin 38: GND
 - Pin 39: +3.3V
 - Pin 40: GND
 - Pin 41: +3.3V
 - Pin 42: GND
 - Pin 43: +3.3V
 - Pin 44: GND
 - Pin 45: +3.3V
 - Pin 46: GND
 - Pin 47: +3.3V
 - Pin 48: GND
 - Pin 49: +3.3V
 - Pin 50: GND
 - Pin 51: +3.3V
 - Pin 52: GND
 - Pin 53: +3.3V
 - Pin 54: GND
 - Pin 55: +3.3V
 - Pin 56: GND
 - Pin 57: +3.3V
 - Pin 58: GND
 - Pin 59: +3.3V
 - Pin 60: GND
 - Pin 61: +3.3V
 - Pin 62: GND
 - Pin 63: +3.3V
 - Pin 64: GND
 - Pin 65: +3.3V
 - Pin 66: GND
 - Pin 67: +3.3V
 - Pin 68: GND
 - Pin 69: +3.3V
 - Pin 70: GND
 - Pin 71: +3.3V
 - Pin 72: GND
 - Pin 73: +3.3V
 - Pin 74: GND
 - Pin 75: +3.3V
 - Pin 76: GND
 - Pin 77: +3.3V
 - Pin 78: GND
 - Pin 79: +3.3V
 - Pin 80: GND
 - Pin 81: +3.3V
 - Pin 82: GND
 - Pin 83: +3.3V
 - Pin 84: GND
 - Pin 85: +3.3V
 - Pin 86: GND
 - Pin 87: +3.3V
 - Pin 88: GND
 - Pin 89: +3.3V
 - Pin 90: GND
 - Pin 91: +3.3V
 - Pin 92: GND
 - Pin 93: +3.3V
 - Pin 94: GND
 - Pin 95: +3.3V
 - Pin 96: GND
 - Pin 97: +3.3V
 - Pin 98: GND
 - Pin 99: +3.3V
 - Pin 100: GND
- GPIOs:**
 - Pin 1: GPIO0
 - Pin 2: GPIO1
 - Pin 3: GPIO2
 - Pin 4: GPIO3
 - Pin 5: GPIO4
 - Pin 6: GPIO5
 - Pin 7: GPIO6
 - Pin 8: GPIO7
 - Pin 9: GPIO8
 - Pin 10: GPIO9
 - Pin 11: GPIO10
 - Pin 12: GPIO11
 - Pin 13: GPIO12
 - Pin 14: GPIO13
 - Pin 15: GPIO14
 - Pin 16: GPIO15
 - Pin 17: GPIO16
 - Pin 18: GPIO17
 - Pin 19: GPIO18
 - Pin 20: GPIO19
 - Pin 21: GPIO20
 - Pin 22: GPIO21
 - Pin 23: GPIO22
 - Pin 24: GPIO23
 - Pin 25: GPIO24
 - Pin 26: GPIO25
 - Pin 27: GPIO26
 - Pin 28: GPIO27
 - Pin 29: GPIO28
 - Pin 30: GPIO29
 - Pin 31: GPIO30
 - Pin 32: GPIO31
 - Pin 33: GPIO32
 - Pin 34: GPIO33
 - Pin 35: GPIO34
 - Pin 36: GPIO35
 - Pin 37: GPIO36
 - Pin 38: GPIO37
 - Pin 39: GPIO38
 - Pin 40: GPIO39
 - Pin 41: GPIO40
 - Pin 42: GPIO41
 - Pin 43: GPIO42
 - Pin 44: GPIO43
 - Pin 45: GPIO44
 - Pin 46: GPIO45
 - Pin 47: GPIO46
 - Pin 48: GPIO47
 - Pin 49: GPIO48
 - Pin 50: GPIO49
 - Pin 51: GPIO50
 - Pin 52: GPIO51
 - Pin 53: GPIO52
 - Pin 54: GPIO53
 - Pin 55: GPIO54
 - Pin 56: GPIO55
 - Pin 57: GPIO56
 - Pin 58: GPIO57
 - Pin 59: GPIO58
 - Pin 60: GPIO59
 - Pin 61: GPIO60
 - Pin 62: GPIO61
 - Pin 63: GPIO62
 - Pin 64: GPIO63
 - Pin 65: GPIO64
 - Pin 66: GPIO65
 - Pin 67: GPIO66
 - Pin 68: GPIO67
 - Pin 69: GPIO68
 - Pin 70: GPIO69
 - Pin 71: GPIO70
 - Pin 72: GPIO71
 - Pin 73: GPIO72
 - Pin 74: GPIO73
 - Pin 75: GPIO74
 - Pin 76: GPIO75
 - Pin 77: GPIO76
 - Pin 78: GPIO77
 - Pin 79: GPIO78
 - Pin 80: GPIO79
 - Pin 81: GPIO80
 - Pin 82: GPIO81
 - Pin 83: GPIO82
 - Pin 84: GPIO83
 - Pin 85: GPIO84
 - Pin 86: GPIO85
 - Pin 87: GPIO86
 - Pin 88: GPIO87
 - Pin 89: GPIO88
 - Pin 90: GPIO89
 - Pin 91: GPIO90
 - Pin 92: GPIO91
 - Pin 93: GPIO92
 - Pin 94: GPIO93
 - Pin 95: GPIO94
 - Pin 96: GPIO95
 - Pin 97: GPIO96
 - Pin 98: GPIO97
 - Pin 99: GPIO98
 - Pin 100: GPIO99
- Other Pins:**
 - Pin 101: THERMAL_PAD
 - Pin 102: NC
 - Pin 103: NC
 - Pin 104: NC
 - Pin 105: NC
 - Pin 106: NC
 - Pin 107: NC
 - Pin 108: NC
 - Pin 109: NC
 - Pin 110: NC
 - Pin 111: NC
 - Pin 112: NC
 - Pin 113: NC
 - Pin 114: NC
 - Pin 115: NC
 - Pin 116: NC
 - Pin 117: NC
 - Pin 118: NC
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 - Pin 140: NC
 - Pin 141: NC
 - Pin 142: NC
 - Pin 143: NC
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 - Pin 145: NC
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 - Pin 147: NC
 - Pin 148: NC
 - Pin 149: NC
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 - Pin 151: NC
 - Pin 152: NC
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 - Pin 162: NC
 - Pin 163: NC
 - Pin 164: NC
 - Pin 165: NC
 - Pin 166: NC
 - Pin 167: NC
 - Pin 168: NC
 - Pin 169: NC
 - Pin 170: NC
 - Pin 171: NC
 - Pin 172: NC
 - Pin 173: NC
 - Pin 174: NC
 - Pin 175: NC
 - Pin 176: NC
 - Pin 177: NC
 - Pin 178: NC
 - Pin 179: NC
 - Pin 180: NC
 - Pin 181: NC
 - Pin 182: NC
 - Pin 183: NC
 - Pin 184: NC
 - Pin 185: NC
 - Pin 186: NC
 - Pin 187: NC
 - Pin

Access Bus

The diagram illustrates the Access Bus connections for the HN1x8 module. It shows a set of 8 green horizontal lines representing the bus. On the left, these lines are labeled with the following signals: ESP_EN, GPIO2_U1RXD, GPIO34_U1RXD, GPIO00_U1RTS, and GPIO39_U1CTS. On the right, the lines are numbered 1 through 8, with a label 'Access_Bus1' above them. Below the numbering, a small red box contains the text 'HN1x8'. Power and ground connections are indicated on the right side: +5V is connected to line 1, +3.3V to line 2, and GND to lines 3 and 4. Lines 5, 6, 7, and 8 are not connected to any specific signal or power source in this diagram.

Signal	Line Number
ESP_EN	1
GPIO2_U1RXD	2
GPIO34_U1RXD	3
GPIO00_U1RTS	4
GPIO39_U1CTS	5
	6
	7
	8

Access_Bus1

HN1x8

[illegible]

VGA1

Red
Green
Blue
ID2/RES
GND
Red_RTN
Green_RTN
Blue_RTN
Key/PWR
GND
ID0/RES
ID1/SDA
HSync
VSync
ID3/SC/L
Shield1
Shield2

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17

R22 430R/1%/R0603
R23 R20R/R0603
R24 430R/1%/R0603
R25 R20R/R0603
R26 430R/1%/R0603
R27 R20R/R0603

GPIO23_Red_1V_LCD_DC
GPIO21_Red_0V_LCD_RST
GPIO19_Green_1V_VSPL_MISO
GPIO18_Green_0V_VSPL_SCK
GPIO15_Blue_1V_VSPL_CS
GPIO14_Blue_0V_2C_SDA
GPIO23_HSync_VSPL_MOSI
GPIO15_VSync_2C_SCL

TVS1 NTC(G0402020284-4P)
TVS2 NTC(G0402020284-4P)
TVS3 NTC(G0402020284-4P)
TVS4 NTC(G0402020284-4P)
TVS5 NTC(G0402020284-4P)

GND GND GND GND GND

HDB15-3.0B-14.5T/VGA15/14.5mm

[illegible]

LCD Connector

The diagram illustrates the pinout for an LCD connector. It consists of 12 pins, numbered 1 through 12. The functions and power levels for each pin are as follows:

Pin Number	Function	Power Level
1	GPI05\Blue_1\VSPL_CS	+3.3V
2	GPI022\Red_1\LCD_DC	GND
3	GPI018\Green_0\VSPL_5CK	
4	GPI023\HSync\VSPL_MISO	
5	GPI019\Green_1\VSPL_MISO	
6	GPI021\Red_0\LCD_RST	
7	GPI015\VSync\2C_SCL	
8	GPI04\Blue_0\2C_SDA	
9		
10		
11	GND	
12	+5V	

The connector is labeled **LCD_CON1** at the top right and **HN1x12** at the bottom right.

Software Selectable Pins		
Interface	Signal	Pin
EMAC	EMAC_MDC_out	Any GPIO
	EMAC_MDI_in	
	EMAC_MDIO_out	
	EMAC_CRDS_out	
I2C	EMAC_CDL_out	Any GPIO
	I2CEXT0_SCL_in	
	I2CEXT0_SDA_in	
	I2CEXT1_SCL_in	
	I2CEXT1_SDA_in	
	I2CEXT0_SCL_out	
	I2CEXT0_SDA_out	
General Purpose SPI	I2CEXT1_SCL_out	Any GPIO
	HSPIQ_in_/out	
	HSPI0_in_/out	
	HSPICLK_in_/out	
	HSPLCS0_in_/out	
	HSPLCS1_out	
	HSPLCS2_out	
	VSPIQ_in_/out	
	VSPIN_in_/out	
	VSPICLK_in_/out	
	VSPIC0_in_/out	
	VSPIC1_out	
	VSPIC2_out	

For more information refer to [esp_wroom_32_datasheet_en.pdf](#).

PS/2 Port 0 -> Keyboard

PS/2 Port 1 -> Mouse

Current Consumption
Approximately: 100mA

PS2_MOUSE1

GPIO27_MOUSE_DAT

GPIO26_MOUSE_CLK

GND

GND

Mounting Holes



MH1

Mounting_Hole_NPTH



MH2

Mounting_Hole_NPTH



MH3

Mounting_Hole_NPTH



MH4

Mounting_Hole_NPTH

[illegible]

UEXT

PC3\UEXT_PWR_EN#

R30 2.2k/0602

R31 220k/0602

C17 100nF/0603

FET2 WPM2015-3/TR

+3.3V

R32 2.2k/0602

R37 2.2k/0602

UEXT1

1 2 3 4 5 6 7 8 9 10

B-R-10-LF (6H10R)

PD5\UTX
PC3\SCL
PA2\SoftSPL_MISO
PD4\SoftSPL_SCK

PD6\URX
PC3\SDA
PA1\SoftSPL_MOSI
PD3\SoftSPL_CS#

Fiducials

 FID1 Fiducial	 FID3 Fiducial	 FID5 Fiducial
 FID2 Fiducial	 FID4 Fiducial	 FID6 Fiducial

